

UNCONDITIONAL DOUBLY EXPONENTIAL UPPER BOUNDS FOR THE BOTTOM OF WINDOWED WEIL QUADRATIC FORMS

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ABSTRACT. For $\lambda > 1$ let $\lambda_{\min}(\lambda)$ be the bottom of the spectrum of Weil’s quadratic form restricted to test functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$. The Riemann Hypothesis is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ . We prove two *upper* bounds that hold without any hypothesis on the zeros of ζ . Theorem A gives $\lambda_{\min}(\lambda) \leq C\lambda^p e^{-2\pi\lambda^2}$, with all constants explicit; for every $\lambda \geq \sqrt{5}$ one may take $\lambda_{\min}(\lambda) \leq 1.563 \times 10^9 \lambda^{16} e^{-2\pi\lambda^2}$. Theorem B improves the exponent to $e^{-4\pi\lambda^2}$ by means of Kaiser–Bessel windows. Both proofs rest on one observation: the image of the summation map $\varphi \mapsto \sum_n \varphi(nu)$ lies in the radical of the Weil form, and cutting such a function off to the window costs only a Gaussian tail. The exponent 4π agrees with the prolate spheroidal heuristics of Connes–Consani–Moscovici and with high-precision computations of $\lambda_{\min}$. We also observe that the prolate quantity $1 - \chi_2(\lambda) \asymp \mu^{9/2} e^{-4\pi\mu}$ reproduces the certified numerical upper bounds of Zhu to within a factor of 20, over 283 orders of magnitude. This numerical observation suggests that the true decay rate has exponent 4π . Theorem A is formalized in Lean 4, on top of a formalization of Weil’s explicit formula; the formal statement depends only on the standard axioms. These results concern upper bounds only. They say nothing about positivity and do not address the Riemann Hypothesis.

Status.

- **Theorem A.** Proved in this paper, with explicit constants established by ball arithmetic. In its form with existential constants it is formalized in Lean 4 (declaration `RhWeil.Concrete.theoremA`). That formalization depends only on `propext`, `Classical.choice` and `Quot.sound`, and its statement was audited against the standard form of the explicit formula.
- **Theorem B.** Proved in this paper; not formalized. The threshold λ_0 is not effective.

This paper has not yet been reviewed by independent human experts.

1. INTRODUCTION

1.1. Weil’s criterion on windows. Weil’s criterion states that the Riemann Hypothesis (RH) holds if and only if a certain quadratic form W is nonnegative on test functions [13, 2]. Connes and Consani [3] and Connes, Consani and Moscovici [4] study the restriction W_λ of W to functions supported in the multiplicative window $[\lambda^{-1}, \lambda]$. Write $\lambda_{\min}(\lambda)$ for the bottom of its spectrum. Then:

- $\lambda_{\min}$ is nonincreasing in λ ;
- RH is equivalent to $\lambda_{\min}(\lambda) \geq 0$ for all λ ;
- numerically $\lambda_{\min}(\lambda)$ is positive but tiny, and the corresponding ground state is close to a prolate spheroidal wave function [3, 4].

Proving positivity window by window therefore requires control of a margin that shrinks very fast with λ . This paper quantifies that shrinking from above, unconditionally.

1.2. Results. Put $\mu = \lambda^2$.

Theorem A. *There are effectively computable constants C, p, λ_0 such that for all $\lambda \geq \lambda_0$*

$$\lambda_{\min}(\lambda) \leq C \lambda^p e^{-2\pi\lambda^2}.$$

More precisely, for $\mu \geq 5$ we have $\lambda_{\min}(\lambda) \leq 2S B(\lambda)^2 / c_0^2$. Here:

- $B(\lambda)$ is an explicit elementary function with $B(\lambda)e^{\pi\mu} = O(\mu^4)$;
- $S = \sum_{\rho} m_{\rho}(1 + |\gamma_{\rho}|^2)^{-1} < 0.0462$;
- $c_0^2 = 0.00894$.

In closed form, for every $\mu \geq 5$, $\lambda_{\min}(\lambda) \leq 1.563 \times 10^9 \mu^8 e^{-2\pi\mu}$.

Theorem B. *There are constants C, p, λ_0 such that for all $\lambda \geq \lambda_0$*

$$\lambda_{\min}(\lambda) \leq C \lambda^p e^{-4\pi\lambda^2}.$$

Both statements are unconditional: no hypothesis on the location of the zeros of ζ is used. In terms of the half-width $a = \log \lambda$ of the window in logarithmic coordinates, $\mu = e^{2a}$, so the bounds decay doubly exponentially in a .

1.3. Relation to previous work.

- **Zhu's conditional bound.** Zhu [15] proves, *assuming RH*, the bound $\lambda^*(L) \leq \exp(-Le^L)$ for windows $[-L, L]$ in logarithmic coordinates, so $L = a$ and $\exp(-Le^L) = \exp(-\frac{1}{2}\sqrt{\mu} \log \mu)$. Theorems A and B remove the hypothesis and replace $\sqrt{\mu} \log \mu$ by $2\pi\mu$ and $4\pi\mu$.
- **Zhu's certified numerics.** Zhu also gives certified *numerical* upper bounds for $\lambda^*(L)$ with $0.5 \leq L \leq 2$, evaluated unconditionally on the geometric side, together with an empirical law $-\ln \lambda^* \simeq 2\pi^2 N(T^*) / \ln N(T^*)$. Section 6 compares these numbers with the prolate quantity $1 - \chi_2(\lambda) \asymp \mu^{9/2} e^{-4\pi\mu}$. The ratio stays between 3 and 20 for $0.5 \leq L \leq 2$, while the numbers themselves range over 283 orders of magnitude. This suggests that $-\ln \lambda_{\min} = 4\pi\mu + O(\log \mu)$, the rate of Theorem B. Zhu's fitted law would instead behave like $2\pi^2\mu$ as $\mu \rightarrow \infty$. We stress that this is an observation about upper bounds only; the true asymptotics can be decided only with lower bounds, which are of the same strength as RH.
- **Positivity on a fixed window.** In earlier work [10] the author proved $W(f) > 0$ on the window $[-\frac{1}{2} \log 5, \frac{1}{2} \log 5]$ with a Lean formalization. The present paper goes in the opposite direction: it bounds from above, unconditionally, the margin that any such window-by-window positivity proof can have.
- **The CCM program.** In [4] two steps remain open in the program:
 - (1) the ground state is simple and even;
 - (2) the ground state is close to the vector k_{λ} built from prolate functions.
 Our results concern neither step directly. The test functions of Theorem B, however, are closely related to k_{λ} .

1.4. Idea of the proof. Let $\mathcal{E}(\varphi)(u) = u^{1/2} \sum_{n \geq 1} \varphi(nu)$. If φ is even and $\varphi(0) = \int \varphi = 0$, then the Mellin transform of $\mathcal{E}(\varphi)$ equals $\zeta(\frac{1}{2} + iz)$ times an entire function. Hence $\mathcal{E}(\varphi)$ lies in the radical of the Weil form: its Fourier transform vanishes at every zero ordinate γ_{ρ} , whether or not ρ is on the critical line. This fact is classical [5, 3].

Now suppose φ is supported in $[-\lambda, \lambda]$. Then $\mathcal{E}(\varphi)$ vanishes for $u \geq \lambda$. Its restriction k to the window differs from $\mathcal{E}(\varphi)$ only by the piece r on $u < \lambda^{-1}$. By Poisson summation this piece is controlled by $\hat{\varphi}$ on $|\xi| \geq \lambda$. Since $\hat{k} = -\hat{r}$ at the zeros, the explicit formula gives

$$|W(k)| \leq 2S (\text{size of } \hat{\varphi} \text{ beyond } \lambda)^2.$$

Theorem A takes φ to be Riemann's function h , whose Fourier transform is itself and decays like $e^{-\pi\xi^2}$, cut off smoothly. Theorem B takes a Kaiser–Bessel window, whose Fourier transform beyond the band edge is smaller by $e^{-2\pi\lambda^2}$.

1.5. Formalization. Theorem A is formalized in Lean 4 [9, 11] as a statement about the geometric side of the explicit formula (Section 7). It builds on the formalization of Weil's explicit formula in the Lean library Zeta23 [14], which is released under the Apache License 2.0.

2. SETTING AND CONVENTIONS

Let $\lambda > 1$ and $a = \log \lambda$. In the variable $y = \log u$, the space $L^2([\lambda^{-1}, \lambda], d^*u)$ becomes $L^2([-a, a], dy)$. We use two Fourier transforms:

$$\hat{f}(z) = \int_{\mathbb{R}} f(y) e^{izy} dy \quad (\text{logarithmic variable}), \quad \hat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(x) e^{2\pi i x \xi} dx \quad (\text{additive variable}).$$

For a nontrivial zero ρ of ζ with multiplicity m_ρ , put $\gamma_\rho = (\rho - \frac{1}{2})/i$. Then $|\operatorname{Im} \gamma_\rho| < \frac{1}{2}$, and the multiset of the γ_ρ is invariant under $\gamma \mapsto -\gamma$.

Weil's quadratic form $W(f) = QW(f, f)$ is given on the geometric side as in [4, (3.19)]. Let $\lambda_{\min}(\lambda)$ be the bottom of the spectrum of its restriction to $L^2([-a, a])$ [4, Thm. 3.6, Cor. 3.7]. Piecewise smooth functions supported in $[-a, a]$ belong to the form domain [4, after (3.27)]. Hence

$$(1) \quad \lambda_{\min}(\lambda) \leq W(f)/\|f\|^2 \quad \text{for every such } f \neq 0.$$

Finally put $S = \sum_{\rho} m_\rho (1 + |\gamma_\rho|^2)^{-1}$.

3. A MASTER PROPOSITION

Definition 3.1 (Hypothesis (H)). A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ satisfies (H) if:

- φ is even, C^1 , and piecewise C^2 ;
- $\operatorname{supp} \varphi \subset [-\lambda, \lambda]$ and $\varphi(\pm\lambda) = 0$;
- $\varphi(v) = O(v^2)$ at 0 and $\int \varphi = 0$;
- there are $\alpha, \alpha' > 1$ with $|\hat{\varphi}(\xi)| \leq C(1 + |\xi|)^{-\alpha}$ and $|\xi \hat{\varphi}'(\xi)| \leq C(1 + |\xi|)^{-\alpha'}$.

Given such φ , put $e(y) = e^{y/2} \sum_{n \geq 1} \varphi(ne^y)$, $k = e \mathbf{1}_{[-a, a]}$ and $r = e \mathbf{1}_{(-\infty, -a]}$. Then $e = 0$ on $[a, \infty)$, so $e = k + r$.

Lemma 3.2 (Explicit formula). *For real, compactly supported, piecewise C^1 functions f with finitely many jumps, $W(f) = \sum_{\rho} m_\rho \hat{f}(\gamma_\rho) \hat{f}(-\gamma_\rho)$, and the sum converges absolutely.*

Proof. For $f \in C_c^\infty$ this is Weil's explicit formula applied to $f * f^*$ [13, 2]. In general, f' is a finite measure, so $|\hat{f}(z)| \leq C_f/(1 + |z|)$ on $|\operatorname{Im} z| \leq \frac{1}{2}$. Mollify f . The zero side converges by dominated convergence, since $\sum m_\rho (1 + |\gamma_\rho|^2)^{-1} < \infty$. The geometric side converges by dominated convergence (archimedean term), pointwise convergence (poles), and finiteness (primes). $\square$

Lemma 3.3 (Poisson summation). *Under (H), $e(y) = e^{-y/2} \sum_{m \geq 1} \hat{\varphi}(me^{-y})$ for all y . Hence $e(y) = O(e^{(\alpha - \frac{1}{2})y})$ and $e'(y) = O(e^{\min(\alpha, \alpha')y - \frac{1}{2}y})$ as $y \rightarrow -\infty$.*

Lemma 3.4 (Radical). *Under (H), $\hat{e}(\pm\gamma_\rho) = 0$ for every nontrivial zero ρ .*

Proof. For $\operatorname{Im} z < -\frac{1}{2}$, Fubini and the substitution $v = ne^y$ give $\hat{e}(z) = \zeta(\frac{1}{2} + iz)\Phi(z)$ with $\Phi(z) = \int_0^\infty \varphi(v) v^{-\frac{1}{2} + iz} dv$. The function Φ is holomorphic on $\operatorname{Im} z < \frac{5}{2}$, and $\hat{e}$ is holomorphic on $\operatorname{Im} z < \alpha - \frac{1}{2}$ by Lemma 3.3. Both sides therefore agree on the connected set $\{\operatorname{Im} z < \alpha - \frac{1}{2}\} \setminus \{-\frac{i}{2}\}$, which contains every $\pm\gamma_\rho$. Finally $\zeta(\rho) = \zeta(1 - \rho) = 0$. $\square$

Proposition 3.5 (Zero-side estimate). *With $A_0 = \int_{-\infty}^{-a} |r| e^{|y|/2}$ and $A_1 = |r(-a)| e^{a/2} + \int_{-\infty}^{-a} |r'| e^{|y|/2}$,*

$$|W(k)| \leq 2(A_0 + A_1)^2 S, \quad S \leq 2\left(1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi\right) < 0.0462.$$

Proof. By Lemmas 3.2 and 3.4, $W(k) = \sum m_\rho \hat{r}(\gamma_\rho) \hat{r}(-\gamma_\rho)$. On the strip we have $|\hat{r}| \leq A_0$, and one integration by parts gives $|z\hat{r}(z)| \leq A_1$. For the bound on S , pair ρ with $1 - \bar{\rho}$ and use the classical identity $\sum_\rho m_\rho \operatorname{Re}(1/\rho) = 1 + \frac{\gamma_E}{2} - \frac{1}{2} \log 4\pi$. $\square$

Lemma 3.6 (Poisson form). *Let $T_0(s) = \int_s^\infty |\hat{\varphi}|$ and $T_1(s) = \int_s^\infty t|\hat{\varphi}'(t)| dt$. Then*

$$A_0 + A_1 \leq B_\varphi(\lambda) := \frac{3}{2} \sum_m \frac{T_0(m\lambda)}{m} + \sum_m \frac{T_1(m\lambda)}{m} + \lambda \sum_m |\hat{\varphi}(m\lambda)|.$$

Proposition 3.7 (Master proposition). *Under (H), if $\|k\|^2 \geq c_0^2$, then $\lambda_{\min}(\lambda) \leq 2S B_\varphi(\lambda)^2 / c_0^2$.*

Proof. By (1), $\lambda_{\min} \leq W(k)/\|k\|^2 \leq |W(k)|/\|k\|^2$. Now apply Proposition 3.5 and Lemma 3.6. The absolute value is needed because, without RH, $W(k)$ could be negative. $\square$

4. PROOF OF THEOREM A

Let $h(x) = \frac{\pi}{2}x^2(2\pi x^2 - 3)e^{-\pi x^2}$. It is a combination of the Hermite functions h_0 and h_4 , so $\hat{h} = h$; moreover $h(0) = \int h = 0$. Put $\delta = \lambda^{-2}$ and $b = \lambda - 2\delta$. Let χ be the C^2 cut-off built from the smoothstep $S_5(t) = 10t^3 - 15t^4 + 6t^5$: it equals 1 on $|x| \leq b$ and 0 on $|x| \geq b + \delta$. Let $\psi = \frac{315}{32}x^2(1 - x^2)_+^3$. Set

$$\varphi_A = h\chi + \varepsilon\psi, \quad \varepsilon = \int h(1 - \chi), \quad g = h(1 - \chi).$$

Then φ_A satisfies (H) with $\alpha = \alpha' = 2$, and $\hat{\varphi}_A = h - \hat{g} + \varepsilon\hat{\psi}$.

Lemma 4.1 (Explicit tails). *Write $h^{(i)} = Q_i e^{-\pi x^2}$ and $U_i(b) = \sum_k |q_{ik}| b^k e^{-\pi b^2}$. Two integrations by parts, the bounds $\sup |S_5^{(j)}| = \frac{15}{8}, \frac{10}{\sqrt{3}}, 60$, and the Gaussian tail bound*

$$\int_s^\infty t^k e^{-\pi t^2} dt \leq \frac{s^{k-1} e^{-\pi s^2}}{2\pi(1 - \frac{k-1}{2\pi s^2})}$$

give $B_{\varphi_A}(\lambda) \leq B(\lambda)$. Here $B(\lambda)$ is an explicit finite expression in $U_i(b)$, Gaussian tails, $\|\psi''\|_1 = 27.194$ and $\|(x\psi)'''\|_1 = 158.23$. Every term of $B(\lambda)e^{\pi\lambda^2}$ is bounded by a polynomial in λ of degree ≤ 8 times $e^{4\pi/\lambda}$.

Lemma 4.2 (Lower bound for $\|k\|$). *For $\mu \geq 5$, $\|k\|^2 \geq \int_{-0.1}^{0.1} |e|^2 \geq 0.00894$.*

Proof. Ball arithmetic (Arb [1]) gives $\int_{-0.1}^{0.1} e_h^2 \geq 0.0093256$, where $e_h(y) = e^{y/2} \sum h(ne^y)$. The perturbation $e - e_h$ is at most 4.5×10^{-3} at $\mu = 5$, and it decreases in μ . Conclude by Minkowski's inequality. $\square$

Proposition 3.7 now gives Theorem A.

For the closed form, put $F(\lambda) = B(\lambda)e^{\pi\mu}/\mu^4$. We show $F \leq 12298$ for all $\mu \geq 5$.

- On $5 \leq \mu \leq 10^4$ we evaluate F in ball arithmetic on 48,882 intervals in λ . The Gaussian factors are combined with $e^{\pi\lambda^2}$ into $e^{-\pi(s^2 - \lambda^2)}$, so that the enclosures do not blow up. The maximum is at $\mu = 5$, where the point value is 12163.95.
- For $\lambda \geq 100$, replace b by λ in the polynomials P_i and bound $e^{\pi(\lambda^2 - b^2)} \leq e^{4\pi/100}$. This turns $B(\lambda)e^{\pi\lambda^2}$ into a Laurent polynomial in λ with nonnegative coefficients and top degree 8. Hence F is nonincreasing there, and $F \leq 57.0$.

With $S \leq 0.046192$ and $c_0^2 \geq 0.00894$, this gives the constant 1.563×10^9 .

As a check of the direction of the inequalities, B exceeds the value of $A_0 + A_1$ computed from the definitions by factors 80–140 for $\mu = 5$ –13.

5. PROOF OF THEOREM B

Let $\nu = 2$ and $\beta = 2\pi\lambda^2$. For $\alpha > -1$ put $\mathfrak{F}_\alpha(s) = \sum_j (s/4)^j / (2^\alpha j! \Gamma(j + \alpha + 1))$. Let

$$\Phi_\beta(t) = (1 - t^2)I_2(\beta\sqrt{1 - t^2})\mathbf{1}_{|t| \leq 1}, \quad \varphi_B(x) = \frac{x^2(x^2 - s)\Phi_\beta(x/\lambda)}{I_2(\beta)},$$

where s is chosen so that $\int \varphi_B = 0$.

Lemma 5.1 (Sonine–Gegenbauer). $\int_{-1}^1 (1 - t^2)^{\nu/2} I_\nu(\beta\sqrt{1 - t^2}) e^{i\omega t} dt = \sqrt{2\pi} \beta^\nu \mathfrak{F}_{\nu+\frac{1}{2}}(\beta^2 - \omega^2)$ [12, Ch. XII].

Lemma 5.2 (Bessel bounds). • $|J_\alpha| \leq 1$ for $\alpha \geq 0$, and $|J_\alpha(w)| \leq (w/2)^\alpha / \Gamma(\alpha + 1)$ [6, 10.14.1, 10.14.4].

- For $\nu \geq \frac{1}{2}$: $I_\nu(z) \leq e^z / \sqrt{2\pi z}$ for $z > 0$, and $I_\nu(z) \geq c_\nu e^z / \sqrt{z}$ for $z \geq 1$, where $c_\nu = e^{-1} 2^{-\nu} / (\sqrt{\pi} \Gamma(\nu + \frac{3}{2}))$.

The last two bounds follow from the Poisson integral for I_ν .

With $\omega = 2\pi\lambda\xi$, the Sonine–Gegenbauer formula expresses $\hat{\varphi}_B$ through $\partial_\omega^j \mathfrak{F}_{5/2}(\beta^2 - \omega^2)$. Beyond the band edge $|\xi| \geq \lambda$ every $\mathfrak{F}$ is evaluated at a nonpositive argument. The prefactor $1/I_2(\beta) \leq c_2^{-1} \sqrt{\beta} e^{-\beta}$ then gives

$$|\hat{\varphi}_B(\xi)| \leq C\lambda^q e^{-2\pi\lambda^2} (\lambda/|\xi|)^{5/2}, \quad |\xi \hat{\varphi}'_B(\xi)| \leq C\lambda^q e^{-2\pi\lambda^2} (\lambda/|\xi|)^{3/2}.$$

So (H) holds with $(\alpha, \alpha') = (\frac{5}{2}, \frac{3}{2})$, and $B_{\varphi_B} \leq C'\lambda^{q+1} e^{-2\pi\lambda^2}$.

For the lower bound on $\|k\|$, dominate $\Phi_\beta(x/\lambda)/I_2(\beta) \leq 36.14 e^{-\pi x^2}$ using the bounds for I_ν . Then $\varphi_B \rightarrow \pi^{-2}h$, and $\|k\|$ is bounded below as in Section 4. This last step is qualitative, so λ_0 is not yet effective. Proposition 3.7 gives Theorem B.

6. NUMERICAL EVIDENCE

6.1. The bottom of the spectrum. We computed $\lambda_{\min}$ in the even sector with a Legendre–Galerkin discretization in ball arithmetic. The dimension N was increased until convergence. We also computed the Rayleigh quotient of the prolate vector k_λ of [4].

μ	N	$\lambda_{\min}$ (even)	$W(k_\lambda)/\ k_\lambda\ ^2$	$1 - \chi_2$ (Fuchs)	$W(k_\lambda)/\ k_\lambda\ ^2(1 - \chi_2)$
5	140	9.27×10^{-18}	1.40×10^{-17}	1.70×10^{-18}	8.2
7	140	6.75×10^{-28}	9.33×10^{-28}	9.42×10^{-29}	9.9
13	260	2.54×10^{-59}	3.49×10^{-59}	2.75×10^{-60}	12.7
15	260	6.04×10^{-70}	8.72×10^{-70}	6.36×10^{-71}	13.7
17	300	1.35×10^{-80}	1.87×10^{-80}	1.36×10^{-81}	13.7
20*	380	1.35×10^{-96}	1.80×10^{-96}	1.20×10^{-97}	15.0

Here $1 - \chi_2(\lambda) \sim \frac{2^{14}}{3} \sqrt{2} \pi^5 \mu^{9/2} e^{-4\pi\mu}$ is the asymptotic prolate eigenvalue defect of [7]. At $\mu = 11$ it agrees with our direct computation of the prolate eigenvalue to within 13%. The data are consistent with $\lambda_{\min} \asymp \mu^{9/2} e^{-4\pi\mu}$, the exponent of Theorem B.

*At $\mu = 20$ the Galerkin value has not fully converged: from $N = 340$ to $N = 380$ it decreased by 30%. Galerkin values are upper bounds for $\lambda_{\min}$.

6.2. Comparison with Zhu’s certified bounds. Zhu [15, Table 3] gives certified upper bounds $\lambda_{\text{Zhu}}^*(L)$ for windows $[-L, L]$, so $\mu = e^{2L}$. Divided by $1 - \chi_2$, they stay within a narrow band over 283 orders of magnitude:

L	μ	λ_{Zhu}^* (upper bound)	$1 - \chi_2$	ratio
0.8	4.95	2.27×10^{-17}	2.95×10^{-18}	7.7
1.2	11.0	9.98×10^{-49}	8.03×10^{-50}	12.4
1.6	24.5	8.61×10^{-121}	5.51×10^{-122}	15.6
2.0	54.6	3.19×10^{-283}	1.66×10^{-284}	19.2

- The ratio grows slowly, like a small power of μ . This is compatible with $-\ln \lambda^* = 4\pi\mu - O(\log \mu)$.
- Zhu fits the empirical law $-\ln \lambda^* \simeq 2\pi^2 N(T^*)/\ln N(T^*)$, with a fitted constant. It behaves like $2\pi^2\mu$ as $\mu \rightarrow \infty$, but in this range it agrees numerically with $4\pi\mu$.
- Both sets of numbers are upper bounds. Deciding the true asymptotics requires lower bounds, which are of RH strength.

7. FORMALIZATION

The Lean statement `RhWeil.Concrete.theoremA` reads

$$\exists C p \lambda_0, \forall \lambda \geq \lambda_0, \exists k : \mathbb{R} \rightarrow \mathbb{C}, k \in C^2, \text{tsupport } k \subseteq [-\log \lambda, \log \lambda], 0 < \|k\|_2^2, \text{ and } \|\text{literatureRHS}(k \star \tilde{k})\| \leq C \lambda^p e^{-2\pi\lambda^2} \|k\|_2^2.$$

Here `literatureRHS` is the name used in [14] for the geometric side of Weil's explicit formula, in the standard form of [8, Thm. 5.12]:

$$\hat{g}(\tfrac{i}{2}) + \hat{g}(-\tfrac{i}{2}) - \sum_n \frac{\Lambda(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) + \frac{1}{2\pi} \int_{\mathbb{R}} \hat{g}(r) \left[\text{Re} \frac{\Gamma'}{\Gamma} \left(\tfrac{1}{4} + \tfrac{ir}{2} \right) - \log \pi \right] dr.$$

For $g = k \star \tilde{k}$ this is the Weil quadratic form $W(k) = QW(k, k)$. As a numerical check of the conventions, take $k(y) = (1 - y^2/a^2)_+^3$ and $\mu = 5$. The geometric side, the zero side (the first 1000 zeros) and a Legendre–Galerkin discretization all give $2.7877938966 \times 10^{-5}$, and they agree to about 10^{-12} .

The formal proof follows a route slightly different from Section 4:

- the window cut-off is smooth, $k = e\eta$, so that no jump occurs;
- Lemma 3.6 is replaced by a sup-type bound: if $|\hat{\varphi}| \leq M/\xi^2$ on $|\xi| \geq \Xi$, then $|\sum_n \varphi(nu)| \leq M\zeta(2)u$ for $u \leq 1/\Xi$;
- the constants are existential.

The radical lemma is proved through Mathlib's Mellin transform and the identity theorem. Poisson summation is taken from Mathlib. The explicit formula is taken from [14]. The lower bound $\|k\|^2 \geq c_0$ uses no numerics: every summand $\varphi((n+1)e^y)$ with $n \geq 1$ is nonnegative. The development has about 3,300 lines. `#printaxioms` reports `propext`, `Classical.choice` and `Quot.sound`.

What is *not* formalized:

- the identification of the Rayleigh quotient with the spectral bottom [4, Thm. 3.6];
- the explicit constant 1.563×10^9 (proved on paper with ball arithmetic);
- Theorem B, which would require a theory of Bessel functions that Mathlib does not yet provide.

8. REMARKS

- (1) **What the theorems say.** Any proof of RH by window-by-window positivity must beat a margin that is, unconditionally, at most $e^{-4\pi\mu + O(\log \mu)}$.
- (2) **The polynomial factor.** It depends on the construction: $\mu^{16.8}$ numerically for φ_B , against about $\mu^{4.5}$ for k_λ .
- (3) **Relation to the CCM program.** The vector k_λ of the program is of the form $\mathcal{E}(\varphi)$ restricted to the window, with φ a prolate function. The mechanism of this paper therefore also bounds $W(k_\lambda)$ from above. Numerically $W(k_\lambda) \approx (8-14)(1 - \chi_2)$.

REPRODUCIBILITY

The Lean development, the numerical scripts and their outputs, and a summary of the verification are available at <https://github.com/moriryota/weil-window-upper-bounds> (release v1.0.0). The Lean development uses Lean v4.33.0-rc2, Mathlib revision 51e6992e, and the Zeta23 formalization at formal-math commit 3635e748. This paper is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0). The code is distributed under the Apache License 2.0.

USE OF GENERATIVE AI

This work was carried out with extensive use of generative AI systems, namely Anthropic Claude and Google Gemini. Under the direction of the author, these systems did the following:

- drafted the proofs and the Lean development;
- implemented and ran the numerical computations;
- reviewed one another’s drafts adversarially;
- searched the literature;
- drafted this manuscript.

The author chose the problem and the strategy, assigned and reviewed the tasks, and takes full responsibility for the content. The correctness of Theorem A in its existential form does not rest on the AI systems, since it was checked by the Lean kernel. The explicit constants rest on the ball-arithmetic computations described above. Theorem B and the correspondence between the prose and the formal statement were checked by the AI systems and have not yet been reviewed by independent human experts. The AI systems are not authors of this paper.

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